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|-----------------|--------------------------------------------------------------------|
| <b>EXP:4</b>    | <b>Virtual Private Server (VPS) Provisioning and Configuration</b> |
| <b>5/8/2026</b> |                                                                    |

## AIM

To provision a Virtual Private Server (VPS) using VirtualBox, install Ubuntu, configure network settings (IP Address, Subnet Mask, Gateway, DNS), and enable SSH remote access.

## OBJECTIVES

- To understand the concept of Virtual Private Server (VPS).
- To create and manage virtual machines using VirtualBox.
- To install Ubuntu Server OS.
- To configure static network settings.
- To enable SSH remote access.

## SOFTWARE REQUIREMENTS

- Oracle VirtualBox
- Ubuntu Server 22.04 LTS ISO (as specified in the lab manual)
- Windows/Linux Host System
- Internet Connection
- Terminal Access

## THEORY

A Virtual Private Server (VPS) is a software-based server created using virtualization technology. It operates independently with its own operating system, storage, memory, and network configuration while sharing the physical hardware with other virtual machines. Virtualization enables efficient resource utilization and forms the foundation of cloud computing platforms. Network configuration parameters:

| Parameter   | Description                                                        |
|-------------|--------------------------------------------------------------------|
| IP Address  | A unique address assigned to a device on a network.                |
| Subnet Mask | Determines the network and host portions of an IP address.         |
| Gateway     | Provides access to networks outside the local network.             |
| DNS         | Converts domain names into IP addresses.                           |
| SSH         | A secure protocol used for remote administration of Linux systems. |

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|  |  |
|--|--|

# PROCEDURE

## Step 1: Create Virtual Machine

| Parameter | Value           |
|-----------|-----------------|
| Name      | VPS-Server      |
| OS Type   | Ubuntu (64-bit) |
| RAM       | 2 GB            |
| CPU       | 2 Cores         |
| Storage   | 20 GB           |

Attach the Ubuntu ISO, start the VM, follow the installation wizard, create a username and password, and complete the installation.

## Step 2: Update Ubuntu

```
sudo apt update
sudo apt upgrade -y
```

## Step 3: Identify Network Interface

```
ip a
```

The practical execution identified the active interface as enp0s3 with the initial NAT address 10.0.2.15/24.

```
eason@eason-VirtualBox:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:48:4c:85 brd ff:ff:ff:ff:ff:ff
    altname enx080027484c85
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute enp0s3
        valid_lft 86296sec preferred_lft 86296sec
    inet6 fd17:625c:f037:2:a00:27ff:fe48:4c85/64 scope global dynamic noprefixroute
        valid_lft 86258sec preferred_lft 14258sec
    inet6 fe80::a00:27ff:fe48:4c85/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
3: ovs-system: <BROADCAST,MULTICAST> mtu 1500 qdisc noop state DOWN group default qlen 1000
    link/ether 46:44:21:9b:c4:fa brd ff:ff:ff:ff:ff:ff
4: br-int: <BROADCAST,MULTICAST> mtu 1442 qdisc noop state DOWN group default qlen 1000
    link/ether 8a:52:e4:f5:71:a7 brd ff:ff:ff:ff:ff:ff
5: virbr0: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc noqueue state DOWN group default qlen 1000
    link/ether 52:54:00:e4:05:05 brd ff:ff:ff:ff:ff:ff
    inet 192.168.122.1/24 brd 192.168.122.255 scope global virbr0
        valid_lft forever preferred_lft forever
6: br-ex: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UNKNOWN group default qlen 1000
    link/ether a2:74:33:e7:71:4c brd ff:ff:ff:ff:ff:ff
    inet 172.24.4.1/24 scope global br-ex
        valid_lft forever preferred_lft forever
    inet6 2001:db8::2/64 scope global
        valid_lft forever preferred_lft forever
    inet6 fe80::a074:33ff:fee7:714c/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever
7: taped5d9c09-e2: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1442 qdisc noqueue master ovs-system state UP group default qlen 1000
    link/ether fa:16:3e:7d:17:b9 brd ff:ff:ff:ff:ff:ff
    inet6 fe80::f816:3eff:fe7d:17b9/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever
8: br-vif15-0001fe: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1442 qdisc noqueue master ovs-system state UP group default qlen 1000
    link/ether 00:0c:29:1f:00:01 brd 00:0c:29:1f:00:0f
    inet6 fd17:625c:f037:2:a00:27ff:fe48:4c85/64 scope global dynamic noprefixroute br-vif15-0001fe
        valid_lft 86258sec preferred_lft 14258sec
    inet6 fe80::a00:27ff:fe48:4c85/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
```

Figure 1. Network interface identification using ip a.

## Step 4: Check Routing Information

```
ip route
```

The VirtualBox NAT gateway was identified as 10.0.2.2 through enp0s3.

```
eason@eason-VirtualBox:~$ ip route
default via 10.0.2.2 dev enp0s3 proto dhcp src 10.0.2.15 metric 100
10.0.2.0/24 dev enp0s3 proto kernel scope link src 10.0.2.15 metric 100
172.24.4.0/24 dev br-ex proto kernel scope link src 172.24.4.1
192.168.122.0/24 dev virbr0 proto kernel scope link src 192.168.122.1 linkdown
```

Figure 2. Routing table showing the VirtualBox NAT gateway.

## Step 5: Configure Static IP

```
sudo nano /etc/netplan/00-installer-config.yaml
```

The following configuration was used for the actual VirtualBox NAT network:

```
network:
  version: 2
  ethernet:
    enp0s3:
      dhcp4: no
      addresses:
        - 10.0.2.100/24
      routes:
        - to: default
          via: 10.0.2.2
      nameservers:
        addresses:
          - 8.8.8.8
          - 8.8.4.4
```

```
eason@eason-VirtualBox:~$ sudo cat /etc/netplan/00-installer-config.yaml
network:
  version: 2
  ethernet:
    enp0s3:
      dhcp4: no
      addresses:
        - 10.0.2.100/24
      routes:
        - to: default
          via: 10.0.2.2
      nameservers:
        addresses:
          - 8.8.8.8
          - 8.8.4.4
```

Figure 4. Saved Netplan configuration verified with `sudo cat`.

## Step 6: Apply Network Configuration

```
sudo netplan apply
```

The command displayed a systemd-networkd reload warning because the desktop environment was not running systemd-networkd in the expected way. Subsequent routing and connectivity checks confirmed that the static configuration was nevertheless active.

## Step 7: Verify Network Configuration

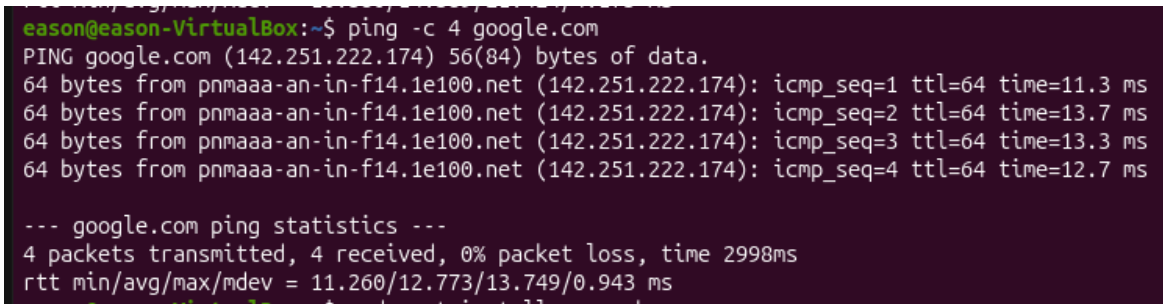
```
ip addr
ip route
cat /etc/resolv.conf
```

The verified practical values were: interface enp0s3, static IP 10.0.2.100/24, default gateway 10.0.2.2, and DNS servers 8.8.8.8 and 8.8.4.4.

## Step 8: Test Connectivity

```
ping -c 4 10.0.2.2
ping -c 4 8.8.8.8
ping -c 4 google.com
```

The gateway, Internet address, and DNS name were all reachable. The practical screenshots showed 4 packets transmitted and 4 received with 0% packet loss for the gateway and 8.8.8.8, and successful DNS resolution of google.com followed by successful replies.



```
eason@eason-VirtualBox:~$ ping -c 4 google.com
PING google.com (142.251.222.174) 56(84) bytes of data.
64 bytes from pnmaaa-an-in-f14.1e100.net (142.251.222.174): icmp_seq=1 ttl=64 time=11.3 ms
64 bytes from pnmaaa-an-in-f14.1e100.net (142.251.222.174): icmp_seq=2 ttl=64 time=13.7 ms
64 bytes from pnmaaa-an-in-f14.1e100.net (142.251.222.174): icmp_seq=3 ttl=64 time=13.3 ms
64 bytes from pnmaaa-an-in-f14.1e100.net (142.251.222.174): icmp_seq=4 ttl=64 time=12.7 ms

--- google.com ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 2998ms
rtt min/avg/max/mdev = 11.260/12.773/13.749/0.943 ms
```

Figure 6. Internet and DNS connectivity verification.

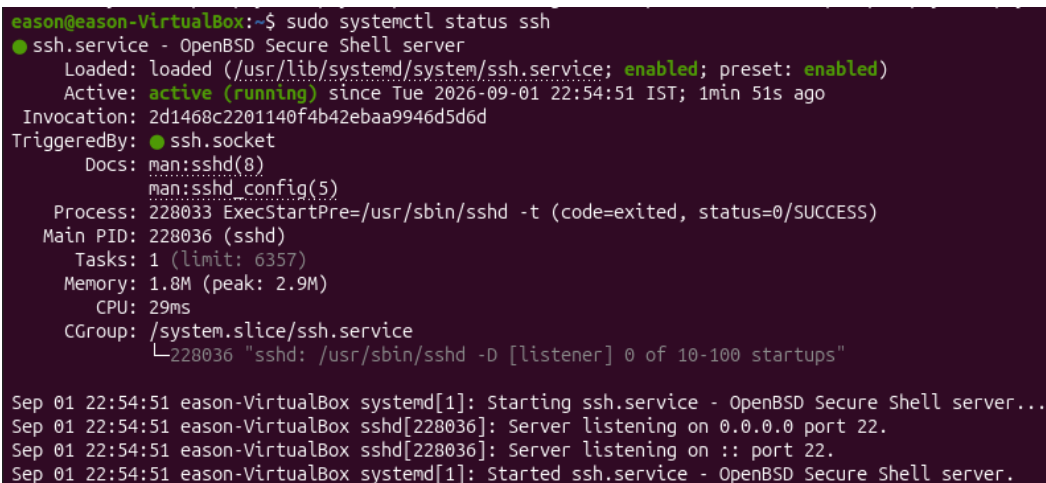
## Step 9: Install OpenSSH Server

```
sudo apt install openssh-server -y
```

## Step 10: Enable and Start SSH Service

```
sudo systemctl enable ssh
sudo systemctl start ssh
sudo systemctl status ssh
```

The SSH service was verified as active (running), enabled, and listening on port 22.



```
eason@eason-VirtualBox:~$ sudo systemctl status ssh
● ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; enabled; preset: enabled)
   Active: active (running) since Tue 2026-09-01 22:54:51 IST; 1min 51s ago
     Invocation: 2d1468c2201140f4b42ebaa9946d5d6d
   TriggeredBy: ● ssh.socket
     Docs: man:sshd(8)
           man:sshd_config(5)
   Process: 228033 ExecStartPre=/usr/sbin/sshd -t (code=exited, status=0/SUCCESS)
   Main PID: 228036 (sshd)
     Tasks: 1 (limit: 6357)
    Memory: 1.8M (peak: 2.9M)
       CPU: 29ms
    CGroup: /system.slice/ssh.service
            └─228036 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

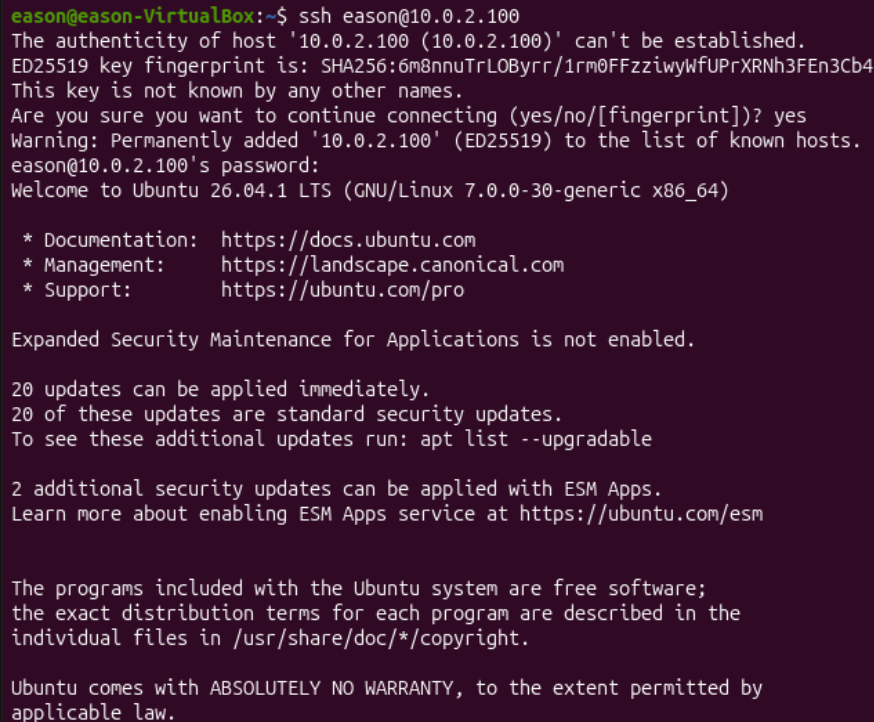
Sep 01 22:54:51 eason-VirtualBox systemd[1]: Starting ssh.service - OpenBSD Secure Shell server...
Sep 01 22:54:51 eason-VirtualBox sshd[228036]: Server listening on 0.0.0.0 port 22.
Sep 01 22:54:51 eason-VirtualBox sshd[228036]: Server listening on :: port 22.
Sep 01 22:54:51 eason-VirtualBox systemd[1]: Started ssh.service - OpenBSD Secure Shell server.
```

Figure 7. OpenSSH service status showing active (running) and port 22 listening.

## Step 11: Remote SSH Login

```
ssh eason@10.0.2.100
```

The SSH connection was successfully established to the static IP address. The Ubuntu welcome message appeared after authentication, demonstrating successful remote login.



```
eason@eason-VirtualBox:~$ ssh eason@10.0.2.100
The authenticity of host '10.0.2.100 (10.0.2.100)' can't be established.
ED25519 key fingerprint is: SHA256:6m8nnuTrLOByrr/1rm0FFzziwyWfUPrXRNh3FEn3Cb4
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '10.0.2.100' (ED25519) to the list of known hosts.
eason@10.0.2.100's password:
Welcome to Ubuntu 26.04.1 LTS (GNU/Linux 7.0.0-30-generic x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

Expanded Security Maintenance for Applications is not enabled.

20 updates can be applied immediately.
20 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

2 additional security updates can be applied with ESM Apps.
Learn more about enabling ESM Apps service at https://ubuntu.com/esm

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.
```

Figure 8. Successful SSH remote login to 10.0.2.100.

## COMMANDS USED

```
sudo apt update
sudo apt upgrade -y
ip a
ip route
sudo nano /etc/netplan/00-installer-config.yaml
sudo cat /etc/netplan/00-installer-config.yaml
sudo netplan apply
ip addr
cat /etc/resolv.conf
ping -c 4 10.0.2.2
ping -c 4 8.8.8.8
ping -c 4 google.com
```

```
sudo apt install openssh-server -y
sudo systemctl enable ssh
sudo systemctl start ssh
sudo systemctl status ssh
ssh eason@10.0.2.100
```

## OBSERVATIONS

| Parameter               | Status                                      |
|-------------------------|---------------------------------------------|
| VM Creation             | Successful                                  |
| Ubuntu Installation     | Successful                                  |
| Network Interface       | enp0s3 identified                           |
| Static IP Configuration | Successful — 10.0.2.100/24                  |
| Subnet Mask             | Successful — /24 (255.255.255.0)            |
| Gateway Configuration   | Successful — 10.0.2.2                       |
| DNS Configuration       | Successful — 8.8.8.8, 8.8.4.4               |
| Gateway Connectivity    | Verified — 0% packet loss                   |
| Internet Connectivity   | Verified — 0% packet loss to 8.8.8.8        |
| DNS Resolution          | Verified — google.com resolved successfully |
| SSH Installation        | Successful                                  |
| SSH Service             | Verified — active (running), port 22        |
| Remote Login            | Successful — ssh eason@10.0.2.100           |

## EXPECTED OUTPUT

- VPS virtual machine created successfully in VirtualBox.
- Ubuntu environment installed and running.
- Static IP, subnet mask, gateway, and DNS configured.
- Internet and DNS connectivity verified.
- OpenSSH Server installed and running.
- SSH remote login completed successfully.

## CONCLUSION

A Virtual Private Server (VPS) environment was successfully created using Oracle VirtualBox. The Ubuntu VM was configured with a static IP address (10.0.2.100/24), gateway (10.0.2.2), and DNS servers (8.8.8.8 and 8.8.4.4). Network connectivity was verified, OpenSSH was installed and enabled, and secure remote access was successfully demonstrated using SSH.

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